Trainings

Initial Check

with Electronically Actuated Injector

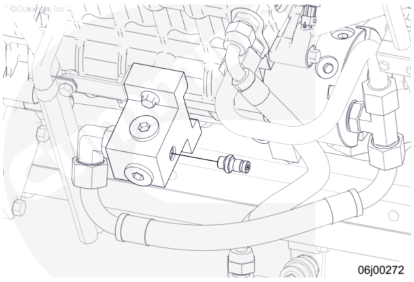
⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

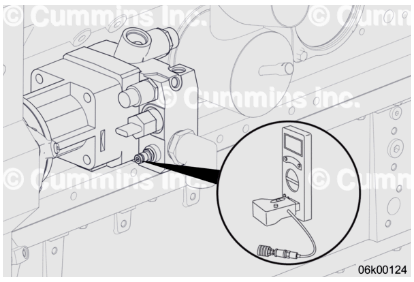
⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

With the fuel supply valve closed, remove the M14 STOR plug from the fuel inlet manifold and install a Compuchek™ fitting, Part Number 3824844, or equivalent.



Connect the vacuum gauge and adapter, Part Number 3164491, or equivalent, and digital multimeter, Part Number 3164488 or 3164489, or equivalent, to the Compuchek™ fitting.

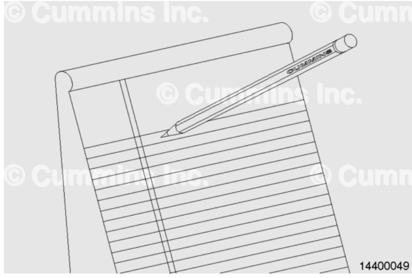


Open the fuel supply valve, and start and operate the engine at high idle.

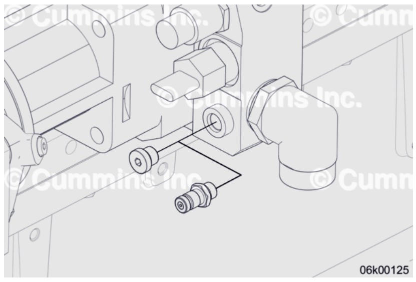
Record the fuel inlet restriction.

The fuel inlet restriction maximum is: 34 kPa [10 in-Hg].

If the restriction is above specifications, measure the Stage 1 fuel filter restriction.



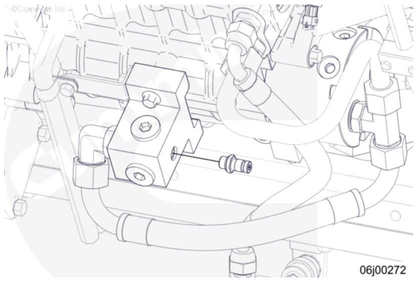
Remove the vacuum gauge and adapter from the fuel inlet manifold block.



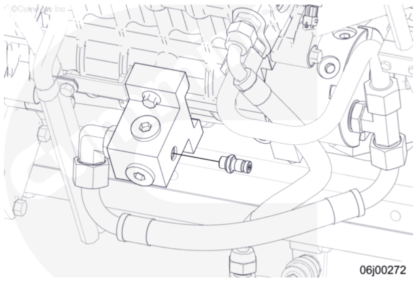
Remove the Compuchek™ fitting and install the threaded o-ring plug into the port in the fuel inlet manifold block.

Torque Value:

Threaded O-ring Plug 29 n•m [21 ft-lb]



For Generator Model DQKAN with Low NOx Emissions Capability

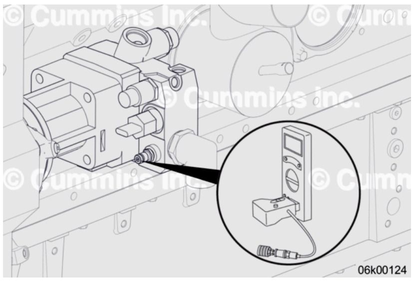


⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

With the fuel supply valve closed, remove the M14 STOR plug from the fuel inlet manifold and install a Compuchek™ fitting, Part Number 3824844, or equivalent.

Connect the Cummins® vacuum gauge and adapter, Part Number 3164491, or equivalent, and Cummins® digital multimeter, Part Number 3164488 or 3164489, or equivalent, to the Compuchek™ fitting.

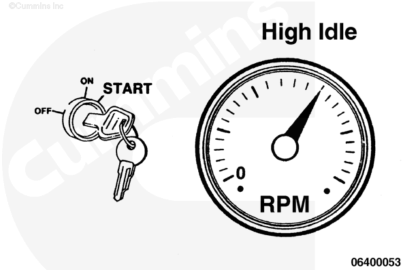


Open the fuel supply valve, and start and operate the engine at high idle.

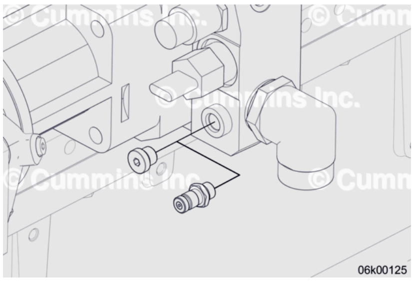
Record the fuel inlet restriction.

The fuel inlet restriction maximum is: 34 kPa [10 in-Hg].

If the restriction is above specifications, measure the Stage 1 fuel filter restriction.

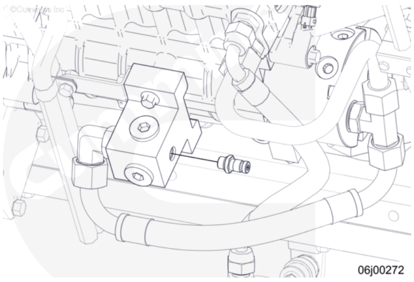


Remove the vacuum gauge and adapter from the fuel inlet manifold block.



Remove the Compuchek™ fitting and install the threaded o-ring plug into the port in the fuel inlet manifold block.

Torque Value:
Threaded O-ring 29 n•m [21 ft-lb]



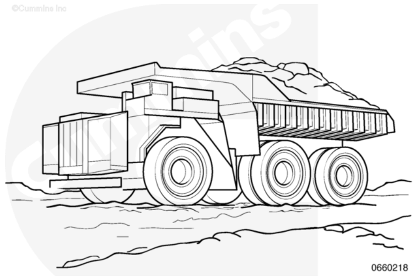
Test

with Mechanically Actuated Injector

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation

to reduce the possibility of severe personal injury or death when working on the fuel system.

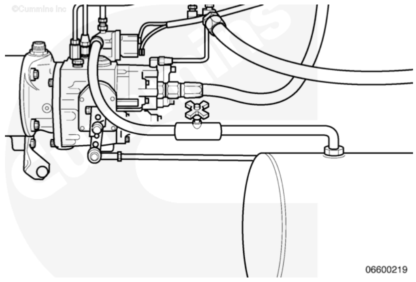


⚠ CAUTION ⚠

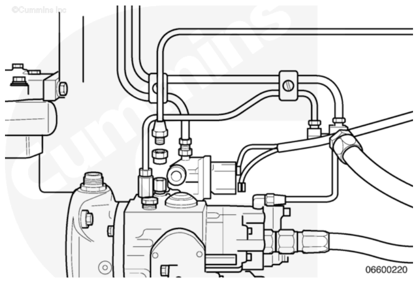
Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

The fuel pump **must** be in the "Full Fuel" position to accurately measure fuel line restriction.

Install a line containing a 7 mm [0.25 in] needle valve between the fuel pump shutoff valve and the fuel tank.



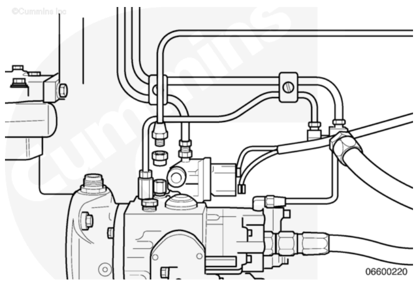
Remove the air/fuel control air supply line from the air intake manifold. Install a plug or a cap in the air manifold hole.



Install the pressure pump, Part Number 3375515, or equivalent.

If the pressure pump is **not** available, install a regulated air pressure hose, with a shutoff valve, to the line.

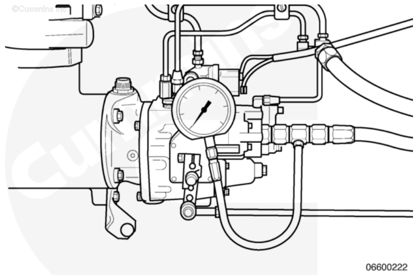
Apply 170 kPa [25 psi] air pressure to the air/fuel control air supply line.



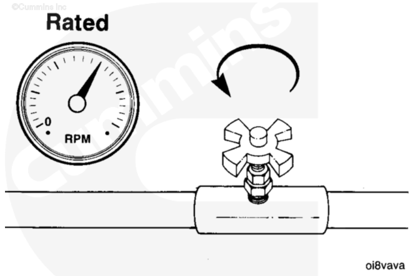
Remove the fuel supply hose to the gear pump or the plug from the rear of the fuel pump mounted to the fuel filter.

Install the adapter as close to the fuel pump as possible. Install a vacuum gauge, Part Number ST-434, or equivalent, to the adapter. The minimum gauge capacity **must** be 760 mm-Hg [30 in-Hg].

Note : The vacuum gauge, Part Number ST434, contains the gauge, hose, and adapter.



With the valve installed, operate the engine at high idle. Slowly open the needle valve until the engine rpm drops to rated rpm. When the engine drops to rated rpm it is at the "Full Fuel" position.



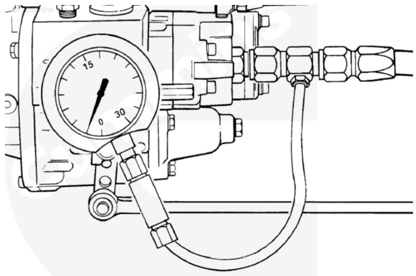
Hold the gauge at the same level as the gear pump.

Note : The gauge will not measure the correct vacuum if it is not held at the same level as the gear pump.

Operate the engine at the “Full Fuel” position.

Observe the reading on the gauge.

The maximum fuel inlet restriction is as follows:



Filter Restriction (Clean)		
kpa		in-hg
14	MAX	4

Filter Restriction (Dirty)		
kpa		in-hg
27	MAX	8

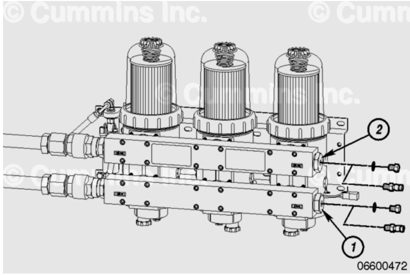
Correct the restriction or replace the fuel filter. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

Measure

Industrial Applications

⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

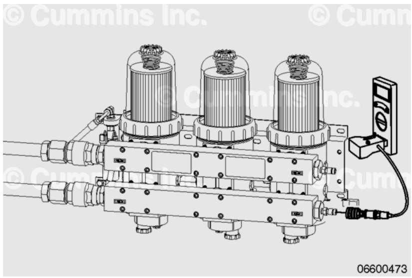


⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

Remove the threaded hex head o-ring plug (1) in the inlet and the threaded hex head o-ring plug (2) in the outlet of Stage 1 filter head manifolds and replace them with Compuchek™ fittings.

Connect a vacuum gauge and adapter to the Compuchek™ fitting in the inlet port.

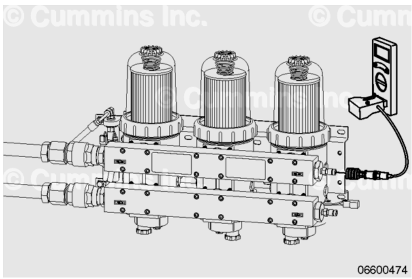


Start and operate the engine at high idle.

Record the Stage 1 inlet restriction. The fuel inlet restriction maximum is 14 kPa [4 in-Hg].

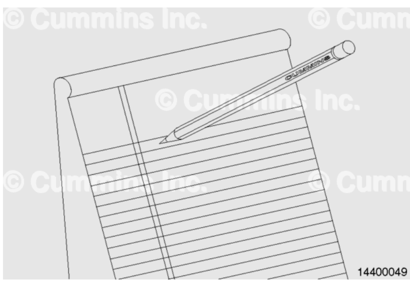
If the restriction is above specifications, inspect the fuel lines. Refer to the original equipment manufacturer (OEM) service manual to determine the source of the high restriction.

Remove the vacuum gauge and adapter from the inlet port and install it on the Compuchek™ fitting in the outlet port.



Start and operate the engine at high idle.

Record the Stage 1 outlet restriction.



Subtract the measurement obtained at the Stage 1 inlet from the measurement obtained at the Stage 1 outlet. This is the Stage 1 restriction.

Example:

- Stage 1 inlet pressure is 3.4 kPa [1 in-Hg]
- Stage 1 outlet pressure is 15.2 kPa [4.5 in-Hg].

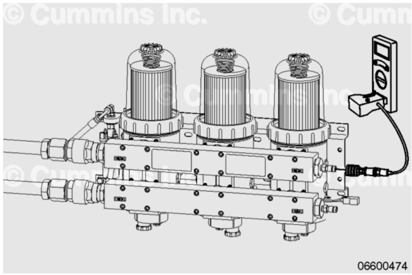
Stage 1 restriction is 15.2 kPa [4.5 in-Hg] minus 3.4 kPa [1 in-Hg] equals 11.8 kPa [3.5 in-Hg].

Stage 1 Filter Restriction		
kpa		in-hg
27.1	MAX	8

If the restriction is above specifications, replace the Stage 1 fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

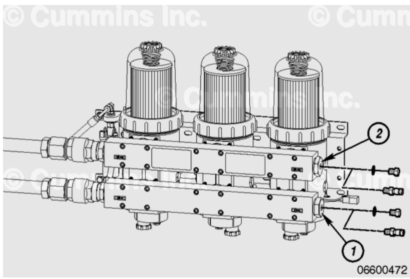
If the restriction is below the maximum specifications, inspect the fuel lines. Refer to the OEM service manual to determine the source of the high restriction.

Remove the vacuum gauge and adapter from the Stage 1 fuel filter head.



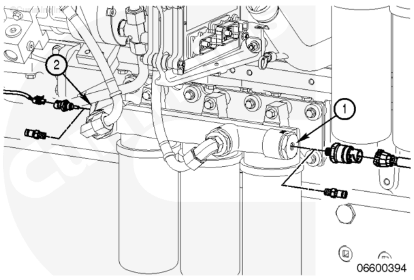
Remove the Compuchek™ fittings and install the threaded hex head o-ring plugs into the inlet port (1) and outlet (2) in the filter head.

Torque Value:
Threaded Hex Head O-ring Plugs 27 n•m [239 in-lb]



Stage 2 Filters

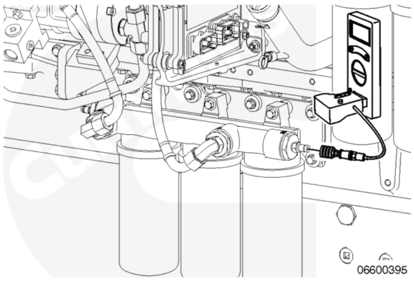
- Remove the Stage 2 filter head inlet pressure sensor (1). See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)
- Remove the Stage 2 filter head temperature sensor (2) See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System)



Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)

- Install Compchek™ fittings into the Stage 2 filter head pressure and temperature sensor ports.

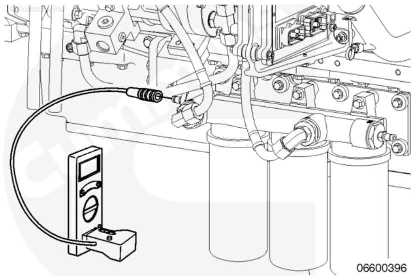
Connect a pressure gauge and adapter to the Compuchek™ fitting in the pressure sensor side of the filter head.



Start and operate the engine at high idle.

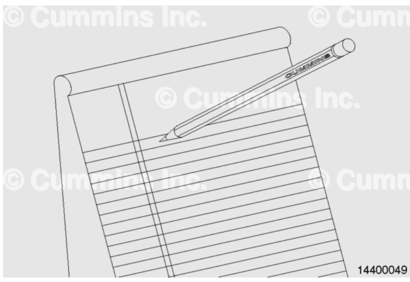
Record the Stage 2 inlet pressure.

Remove the pressure gauge and adapter from the pressure sensor side fitting and install the pressure gauge and adapter on the fitting on the temperature sensor side of the filter head.



Start and operate the engine at high idle.

Record the Stage 2 outlet restriction.



Subtract the measurement obtained at the Stage 2 inlet from the measurement obtained at the Stage 2 outlet. This is the Stage 2 filter restriction.

Example:

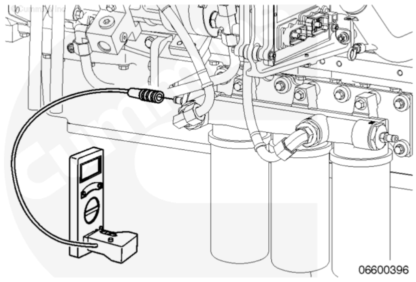
- Stage 2 inlet pressure is 720.5 kPa [104.5 psi]
- Stage 2 outlet pressure is 703.0 kPa [102.0 psi].

Stage 2 restriction is 720.5 kPa [104.5 psi] minus 703.0 kPa [102.0 psi] equals 17.5 kPa [2.5 psi].

Stage 2 Filter Restriction Limts		
kpa		psi
300	MAX	43.5

If the restriction is above specifications, replace the Stage 2 fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

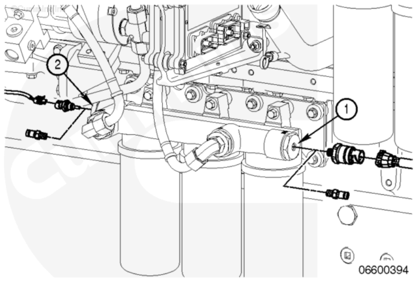
Remove the pressure gauge and adapter from the Stage 2 filter head.



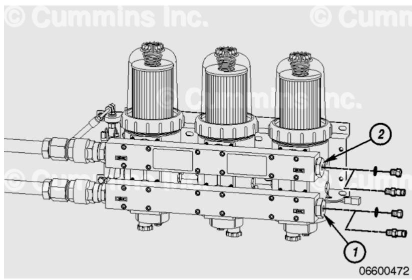
Remove the Compuchek™ fittings from the Stage 2 filter head.

Install the pressure sensor (1) and temperature sensor (2) into the filter head.

See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)



For Generator Model DQKAN with Low NOx Emissions Capability



⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

Stage 1 Filters

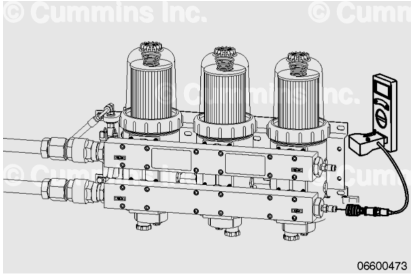
- Remove the threaded hex head o-ring plug (1) in the inlet and the threaded hex head o-ring plug (2) in the outlet of Stage 1 filter head manifolds and replace them with Compuchek™ fittings.

Connect a vacuum gauge and adapter to the Compuchek™ fitting in the inlet port.

Start and operate the engine at high idle.

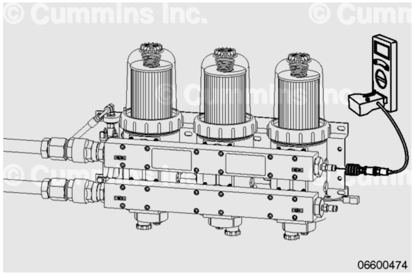
Record the Stage 1 inlet restriction. The fuel inlet restriction maximum is 14 kPa [4 in-Hg].

If the restriction is above specifications, inspect the fuel lines. Reference the equipment manufacturer service information to determine the source of the high restriction.



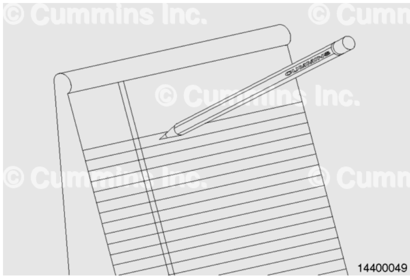
Remove the vacuum gauge and adapter from the inlet port and install it on the Compuchek™ fitting in the outlet port.

Start and operate the engine at high idle.

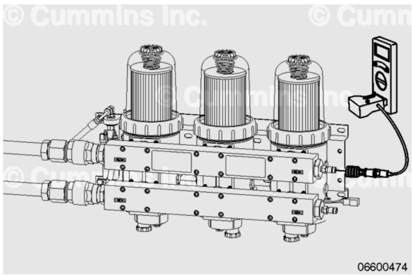


If the restriction is above specifications, replace the Stage 1 fuel filters. Refer to Procedure 006-075 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-075-tr.html)

If the restriction is below the maximum specifications, inspect the fuel lines. See equipment manufacturer service information to determine the source of the high restriction.

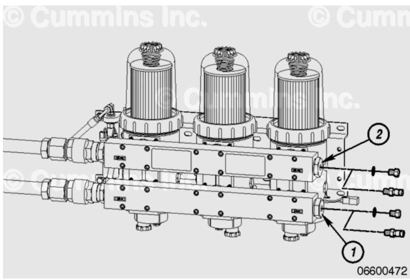


Remove the vacuum gauge and adapter from the Stage 1 fuel filter head.



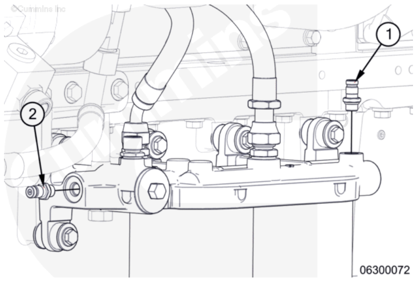
Remove the Compuchek™ fittings and install the threaded hex head o-ring plugs into the inlet port (1) and outlet (2) in the filter head.

Torque Value:
Threaded Hex Head O-ring Plugs 27 n•m [239 in-lb]



Stage 2 Filters

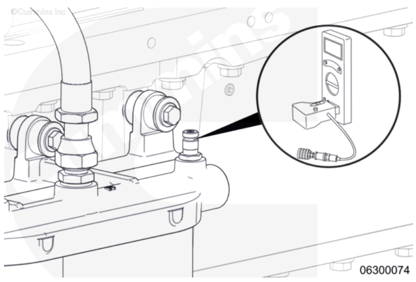
- Remove the Stage 2 filter head inlet pressure sensor (1). Use the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-398 in Section 19 (/qs3/pubsys2/xml/en/procedures/122/122-019-398.html).
- Remove the Stage 2 filter head temperature sensor (2). Use the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-119.html)
- Install Compchek™ fittings into the Stage 2 filter head pressure and temperature sensor ports.



Connect a pressure gauge and adapter to the Compuchek™ fitting in the pressure sensor side of the filter head.

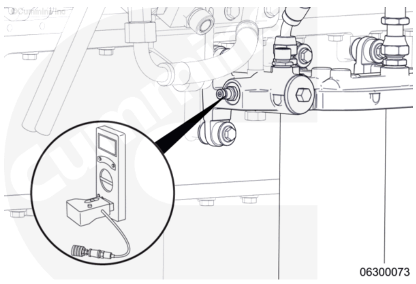
Start and operate the engine at high idle.

Record the Stage 2 inlet pressure.



Remove the pressure gauge and adapter from the pressure sensor side fitting and install the pressure gauge and adapter on the fitting on the temperature sensor side of the filter head.

Start and operate the engine at high idle.



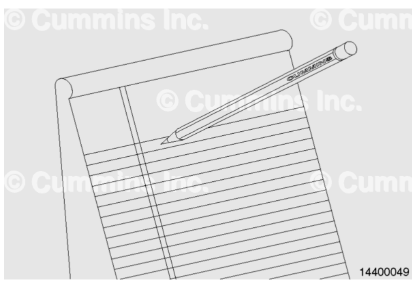
Record the Stage 2 outlet restriction.

Subtract the measurement obtained at the Stage 2 inlet from the measurement obtained at the Stage 2 outlet. This is the Stage 2 filter restriction.

Example:

- Stage 2 inlet pressure is 720.5 kPa [104.5 psi]
- Stage 2 outlet pressure is 703.0 kPa [102.0 psi].

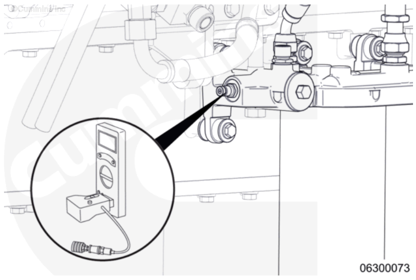
Stage 2 restriction is 720.5 kPa [104.5 psi] minus 703.0 kPa [102.0 psi] equals 17.5 kPa [2.5 psi].



Stage 2 Filter Restriction Limits		
kpa		psi
300	MAX	43.5

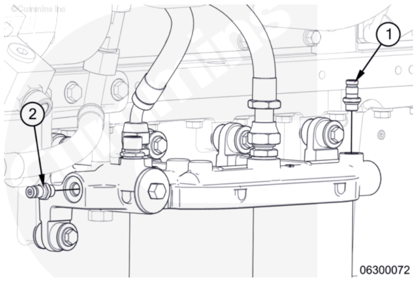
If the restriction is above specifications, replace the Stage 2 fuel filters. Refer to Procedure 006-076 (/qs3/pubsys2/xml/en/procedures/56/56-006-076-tr.html) in Section 6.

Remove the pressure gauge and adapter from the Stage 2 filter head.



Remove the Compuchek™ fittings from the Stage 2 filter head.

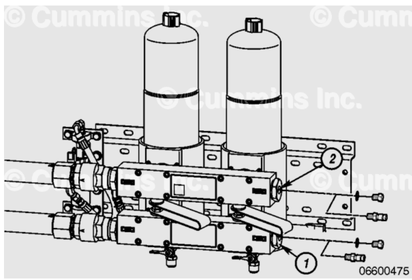
Install the pressure sensor (1) and temperature sensor (2) into the filter head. Use the QSK38, QSK50, and QSK60 CM2150 Electronic Control System Troubleshooting and Repair Manual, Bulletin 4022102. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-119.html)



Marine Applications

Stage 1 Filters

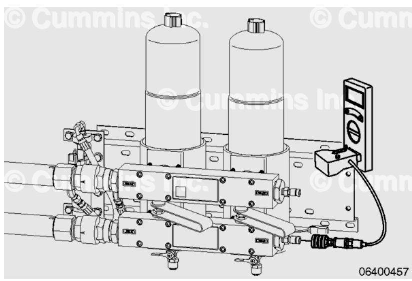
Remove the threaded hex head o-ring plug (1) in the inlet and the threaded hex o-ring plug (2) in the outlet of the Stage 1 filter head manifolds and replace them with Compuchek™ fittings.



Connect a vacuum gauge and adapter to the Compuchek™ fitting in the inlet port.

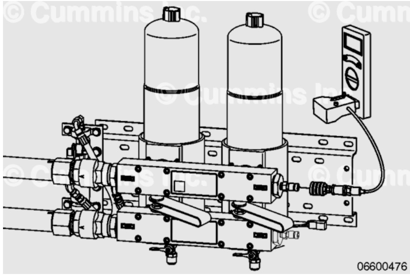
Start and operate the engine at high idle.

Record the Stage 1 inlet restriction.



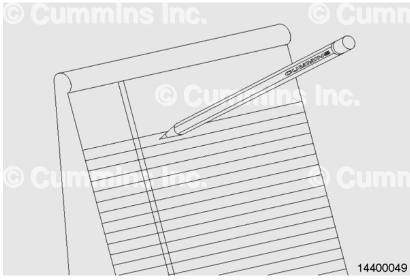
Remove the vacuum gauge and adapter from the inlet port and install it on the Compuchek™ fitting in the outlet port.

Start and operate the engine at high idle.



Record the Stage 1 outlet restriction.

Subtract the measurement obtained at the Stage 1 inlet from the measurement obtained at the Stage 1 outlet. This is the Stage 1 restriction.



Example:

- Stage 1 inlet pressure is 3.4 kPa [1 in-Hg]
- Stage 1 outlet pressure is 15.2 kPa [4.5 in-Hg].

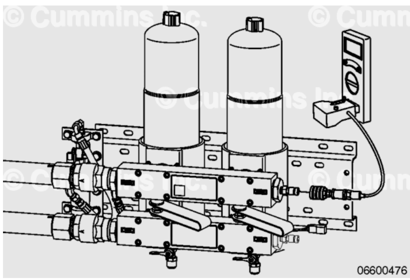
Stage 1 restriction is 15.2 kPa [4.5 in-Hg] minus 3.4 kPa [1 in-Hg] equals 11.8 kPa [3.5 in-Hg].

Stage 1 Filter Restriction		
kpa		in-hg
27.1	MAX	8

If the restriction is above specifications, replace the Stage 1 fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

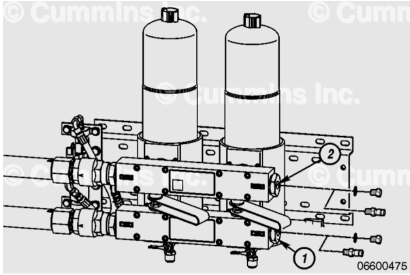
If the restriction is below the maximum specifications, inspect the fuel lines. Refer to the OEM service manual to determine the source of the high restriction.

Remove the vacuum gauge and adapter fitting from the Stage 1 fuel filter head.



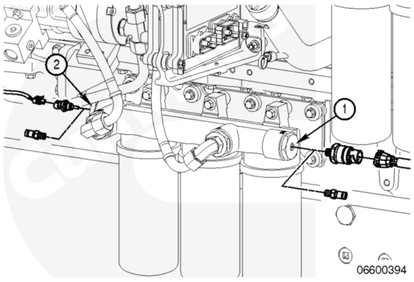
Remove the Compuchek™ fittings and install the threaded hex head o-ring plugs into the inlet port (1), and outlet port (2) in the filter head.

Torque Value:
Threaded Hex Head O-ring Plugs 27 n•m [239 in-lb]



Stage 2 Filters

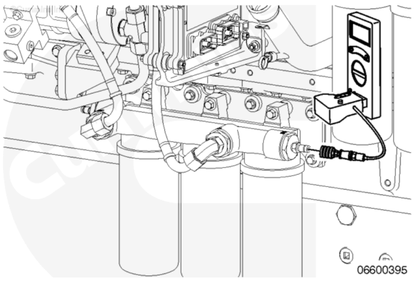
- Remove the Stage 2 filter head inlet pressure sensor (1). See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)
- Remove the Stage 2 filter head temperature sensor (2) See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)
- Install Compchek™ fittings into the Stage 2 filter head pressure and temperature sensor ports.



Connect a pressure gauge and adapter to the Compuchek™ fitting in the pressure sensor side of the filter head.

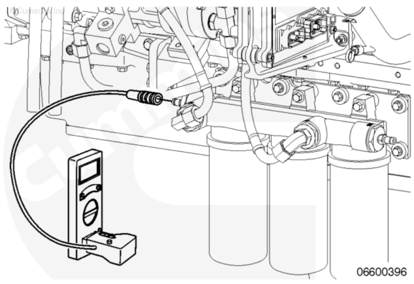
Start and operate the engine at high idle.

Record the Stage 2 inlet pressure.



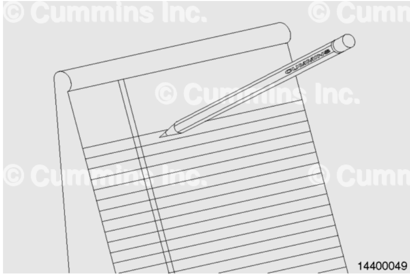
Remove the pressure gauge and adapter from the pressure sensor side fitting and install the pressure gauge and adapter on the fitting on the temperature sensor side of the filter head.

Start and operate the engine at high idle.



Record the Stage 2 outlet pressure.

Subtract the measurement obtained at the Stage 2 outlet from the measurement obtained at the Stage 2 inlet. This is the Stage 2 filter restriction.



Example:

- Stage 2 inlet pressure is 720.5 kPa [104.5 psi]
- Stage 2 outlet pressure is 703.0 kPa [102.0 psi].

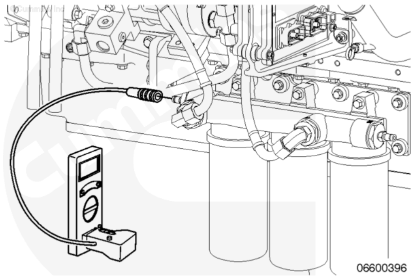
Stage 2 restriction is 720.5 kPa [104.5 psi] minus 703.0 kPa [102.0 psi] equals 17.5 kPa [2.5 psi].

Stage 2 Filter Restriction Limits

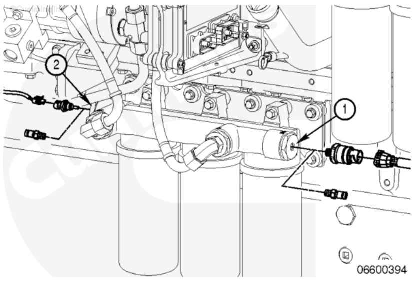
kpa		psi
300	MAX	43.5

If the restriction is above specifications, replace the Stage 2 fuel filters. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-015-tr.html)

Remove the pressure gauge and adapter from the Stage 2 filter head.



Remove the Compuchek™ fittings from the Stage 2 filter head.



Install the pressure sensor (1) into the filter head. See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)

Install the temperature sensor (2) into the filter head. See the following procedure in the QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System) Troubleshooting and Repair Manual, Bulletin 4021533. Refer to Procedure 019-119 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-119.html)

Last Modified: 01-Dec-2021